RACE Regular Analysis Through Continuous Exercise

Kinematics : RACE-01

1. An automobile, travelling at 40 km/h, can be stopped at a distance of 40 m by applying brakes. If the same automobile is travelling at 80 km/h, the minimum stopping distance, in meters, is (Assume no skidding)

(A) 100 m (B) 75 m (C) 160 m (D) 150 m

PKN307

2. A particle is moving in x axis according to relation $x = (4t - t^2 - 4)m$:

(A) Maximum x coordinate of particle is $4m$.

(B) Magnitude of average velocity is equal to average speed, for time interval $t = 0s$ to $t = 2s$.

(C) Average acceleration is equal to instantaneous acceleration during interval $t = 0$ to $t = 2s$.

(D) Distance travelled in interval $t = 0$ to $t = 4$s is 8 m.

PKN308

3. A particle is moving eastwards with a velocity of 5 ms^{-1}. In 10 seconds, the velocity changes to 5 ms^{-1} northward. The average acceleration in this time is

(A) $\frac{1}{2}ms^{-2}$ towards North (B) $\frac{1}{\sqrt{2}}ms^{-2}$ towards North-East

(C) $\frac{1}{\sqrt{2}}ms^{-2}$ towards North-West (D) Zero

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PKN309

4. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is

(A) $v_0 + g/2 + f$ (B) $v_0 + 2g + 3f$ (C) $v_0 + g/2 + f/3$ (D) $v_0 + g + f$

PKN310

5. In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed 'v' more than that of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then 'v' is equal to

(A) $\frac{a_1 + a_2}{2}t$ (B) $\sqrt{2a_1a_2}t$ (C) $\frac{2a_1a_2}{a_1 + a_2}t$ (D) $\sqrt{a_1a_2}t$

PKN311

Paragraph for Question No. 6 to 8

A particle moves along a straight line such that its position x, relative to a fixed point O, on the path, changes with time t as $x = t^2(t - 1)$. Here t is in seconds and x is in meters.

6. The acceleration of the particle at the instant when the particle is at O, is

(A) 4 ms^{-2} (B) -2 ms^{-2} (C) Both (D) None of these

PKN312

For More Join: @IITJEE_Advanced

7. Its distance from O when its velocity is zero, is

(A) 0 (B) $\frac{4}{27}m$ (C) Both (D) None of these

PKN313

8. Its distance from O when its acceleration is zero, is

(A) $\frac{2}{27}m$ (B) 0 (C) Both (D) None of these

PKN314

9. The position of a particle as a function of time t, is given by

$x(t) = at + bt^2 - ct^3$

where a, b and c are constants. When the particle attains zero acceleration, then its velocity will be:

(A) $a + \frac{b^2}{4c}$ (B) $a + \frac{b^2}{c}$ (C) $a + \frac{b^2}{2c}$ (D) $a + \frac{b^2}{3c}$

PKN315

10. An object, moving with a speed of 6.25 m/s, is decelerated at a rate given by

$\frac{dv}{dt} = -2.5\sqrt{v}$

where v is the instantaneous speed. The time taken by the object, to come to rest, would be :-

(A) $4s$ (B) $8s$ (C) $1s$ (D) $2s$

PKN316

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	BCD	C	C	D	C	C	A	D	D

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Kinematics : RACE-02

Paragraph for Question No. 1 to 3

A particle starting from rest has a constant acceleration of 4 ms^{-2} for 4*s*. It then retards uniformly for next 8*s* and comes to rest.

1. Average speed of the particle during the motion is:-

(A) 16 ms^{-1} (B) 8 ms^{-1} (C) 24 ms^{-1} (D) None of these

PKN317

2. Magnitude of average velocity of the particle for a time interval from $t = 0$ to $t = 8$ *s* is :-

(A) 10 ms^{-1} (B) 8 ms^{-1} (C) 12 ms^{-1} (D) None of these

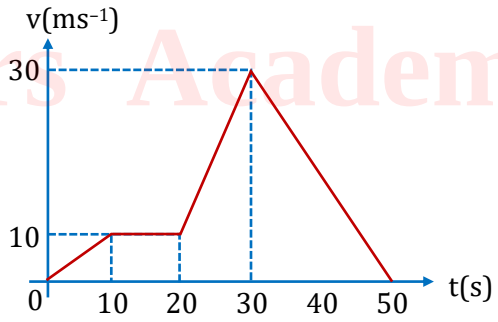
PKN318

3. Magnitude of average acceleration of the particle for a time interval from $t = 0$ to $t = 8$ *s* is

(A) 1 ms^{-2} (B) 2 ms^{-2} (C) 4 ms^{-2} (D) None of these

PKN319

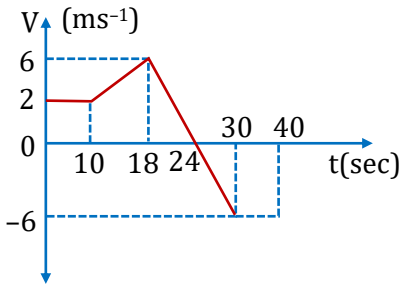
4. Write down maximum acceleration (in ms^{-2}) for shown velocity – time graph of a moving body.



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PKN320

5. A particle moves in a straight line with the velocity as shown in the Figure. At $t = 0$, $x = -16m$.



(A) The maximum value of the position coordinate of the particle is 54 *m*.

(B) The maximum value of the position coordinate of the particle is 36 *m*.

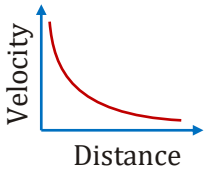
(C) The particle is at the position of 36 *m* at $t = 18$ sec.

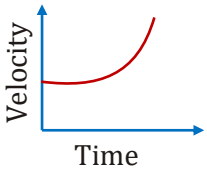
(D) The particle is at the position of 36 *m* at 30 sec.

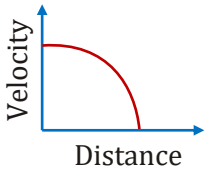
PKN321

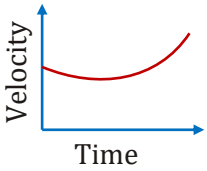
For More Join: @IITJEE_Advanced

6. Which graph corresponds to an object moving with a constant negative acceleration and a positive velocity?

(A) 

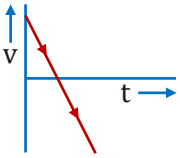
(B) 

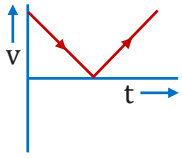
(C) 

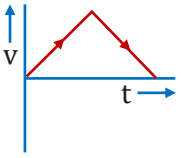
(D) 

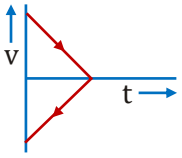
PKN322

7. A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity *vs* time?

(A) 

(B) 

(C) 

(D) 

PKN323

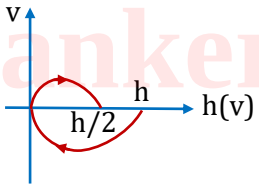
8. A person climbs up a stalled escalator in 60*s*. If standing on the same but escalator running with constant velocity he takes 40*s*. How much time is taken by the person to walk up the moving escalator?

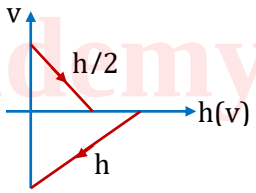
(A) 27*s* (B) 45*s* (C) 37*s* (D) 24*s*

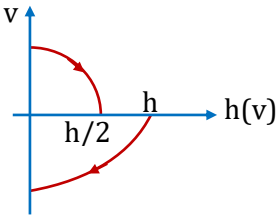
PKN324

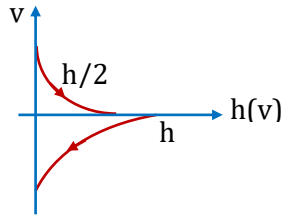
9. A Tennis ball is released from a height *h* and after freely falling on a wooden floor it rebounds and reaches height $\frac{h}{2}$. The velocity versus height of the ball during its motion may be represented graphically by: (Graph are drawn schematically and on not to scale)

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(A) 

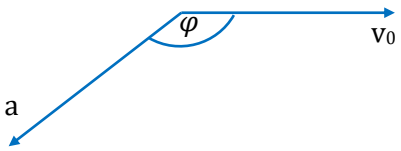
(B) 

(C) 

(D) 

PKN325

10. The figure shows the velocity and the acceleration of a point like body at the initial moment of its motion. The direction and the absolute value of the acceleration remain constant. Find the time in seconds when the speed becomes minimum.
(Given : $a = 4\ m/s^2$, $v_0 = 40\ m/s$, $\phi = 143°$)



PKN326

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	A	2	ACD	C	A	D	C	8

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Kinematics : RACE-03

1. A particle is moving in three dimensions. Its position vector is given by

$\vec{r} = 6\hat{i} + (3+4t)\hat{j} - (3+2t-t^2)\hat{k}$

Distances are in meters, and the time, t, in seconds.

(a) What is the velocity vector at t = +3?

(b) What is the speed (in m/sec) at t = +3?

(c) What is the acceleration vector and what is its magnitude (in m/sec^2) at t = +3?

PKN327

2. A particle is projected with a velocity $\vec{v}_0 = a\hat{i} + b\hat{j}$ from ground with y-axis as vertical and it lands on ground.

Column-I	Column-II
(A) $\frac{v_x}{v_y}$ at $t = \frac{b}{2g}$	(P) $\frac{a}{\sqrt{a^2+b^2}}$
(B) h_{max}	(Q) $\frac{2a}{b}$
(C) R (Horizontal)	(R) $\frac{b^2}{2g}$
(D) $\frac{v_{av}}{v_0}$ where v_{av} = Average velocity for entire journey	(S) $\frac{2ab}{g}$
	(T) $\frac{a}{b}$

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PKN328

3. A car is traveling on a straight level track with a uniform speed of 25 m/s. When the car is moving away from a gun at the same level at a distance of 1500 m the gun is fired at an angle of 45°. Find the distance of the car (kilometers) from the gun when it is hit.

PKN329

4. A projectile is thrown with some initial velocity u at an angle θ to the horizontal. Its speed when it is at the highest point is $\sqrt{\frac{2}{5}}$ times the speed v when it is at height half of the maximum height. Then ratio $\frac{v}{u} = \sqrt{\frac{n}{8}}$. Find the value of n.

PKN330

5. A body is projected at t = 0 with a velocity 10 ms^{-1} at an angle of 60° with the horizontal. The radius of curvature of its trajectory at t = 1s is R. Neglecting air resistance and taking acceleration due to gravity g = 10 ms^{-2}, the value of R is

(A) 2.5m (B) 10.3m (C) 2.8m (D) 5.1m

PKN331

For More Join: @IITJEE_Advanced

6. A shell is fired from a fixed artillery gun with an initial speed u such that it hits the target on the ground at a distance R from it. If t_1 and t_2 are the values of the time taken by it to hit the target in two possible ways, the product t_1t_2 is:

(A) R/g (B) $R/4g$ (C) $2R/g$ (D) $R/2g$

PKN332

7. Two particles are projected from the same point with the same speed u such that they have the same range R, but different maximum heights, h_1 and h_2. Which of the following is correct ?

(A) $R^2 = 2\,h_1h_2$ (B) $R^2 = 16h_1h_2$ (C) $R^2 = 4\,h_1h_2$ (D) $R^2 = h_1h_2$

PKN333

8. A ball is thrown on the level ground such that it just crosses two poles of same height 1 second and 3 seconds after the projection. The distance between the poles is 20 m. Find the ratio of vertical component of velocity of projection and horizontal component of velocity of projection.

PKN334

9. A ball was thrown from a point A with an initial velocity of 30 m/sec at an angle of 45° with the horizontal on a planet Newtonian where value of acceleration due to gravity was $1/5^{th}$ of the value on earth. A wind was blowing in horizontal direction (in the plane of motion of ball) which cause a constant acceleration of $2m/s^2$. The range of projectile (in m) is given by

(A) 450 (B) zero (C) 900 (D) 300

PKN335

10. A water fountain on the ground sprinkles water all around it. If the speed of water coming out of the fountain is v, the total area around the fountain that gets wet is :-

(A) $\frac{\pi}{2}\frac{v^4}{g^2}$ (B) $\pi\frac{v^2}{g^2}$ (C) $\pi\frac{v^2}{g}$ (D) $\pi\frac{v^4}{g^2}$

PKN336

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	(a) $\vec{v} = 4\hat{j} + 4\hat{k}$ (m/s) (b) $4\sqrt{2}$ m/s (c) 2 m/s^2	(A-Q), (B-R), (C-S), (D-P)	2	5	C	C	B	2	BC	D

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RACE Regular Analysis Through Continuous Exercise

Kinematics : RACE-04

1. The position of a projectile launched from the origin at $t = 0$ is given by $\vec{r} = \left(40\hat{i} + 50\hat{j}\right)$ m at $t = 2s$. If the projectile was launched at an angle θ from the horizontal, then θ is (Take $g = 10\ ms^{-2}$).

(A) $\tan^{-1}\frac{4}{5}$ (B) $\tan^{-1}\frac{3}{2}$ (C) $\tan^{-1}\frac{2}{3}$ (D) $\tan^{-1}\frac{7}{4}$

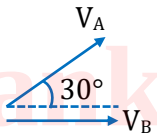
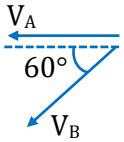
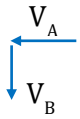
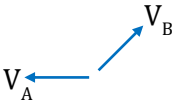
PKN337

2. A ball projected from ground at an angle of 45° just clears a wall in front. If point of projection is $4m$ from the foot of wall and ball strikes the ground at a distance of $6m$ on the other side of the wall, the height of the wall is :

(A) 4.4 m (B) 2.4 m (C) 1.7 m (D) 3.6 m

PKN338

3. Column-I gives the direction of velocity vectors of the two particles A and B. Column-II gives the possible directions of vectors of relative velocity.

Column-I	**Column-II**
(A)  Velocity of B with respect to A	(P) 
(B)  Velocity of A with respect to B	(Q) 
(C)  Velocity of B with respect to A	(R) 
(D)  Velocity of A with respect to B	(S)  (T) 

PKN339

4. Two cars are moving towards each other with velocity 4 m/s and 8 m/s respectively from two different points in straight line, 1200 m apart simultaneously, find time after which both cars meet:

(A) 300 sec (B) 150 sec (C) 100 sec (D) None of these

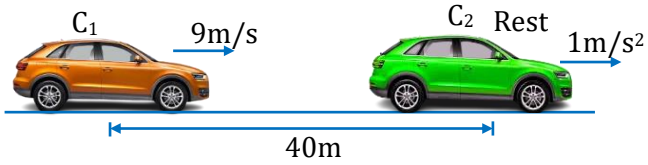
PKN340

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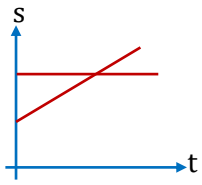
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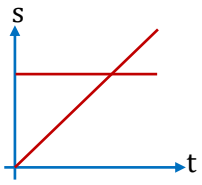
Paragraph for Question No. 5 to 7

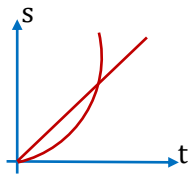
Two cars travelling on a straight road. The car C_1 is going at a constant speed of 9 m/s. 40 m ahead of C_1 is car C_2 at rest, but starts moving with an acceleration of 1 m/s^2.

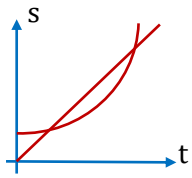


5. The graph correctly representing the position of two cars with time w.r.t. initial position of C_1.

(A) 

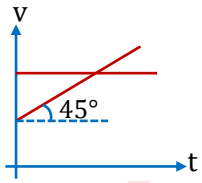
(B) 

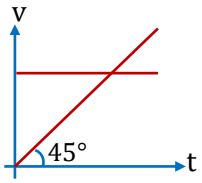
(C) 

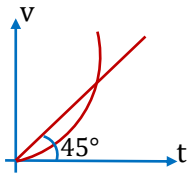
(D) 

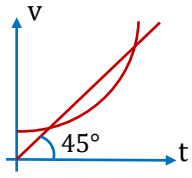
PKN341

6. The graph correctly representing the velocity of the two cars with time

(A) 

(B) 

(C) 

(D) 

PKN342

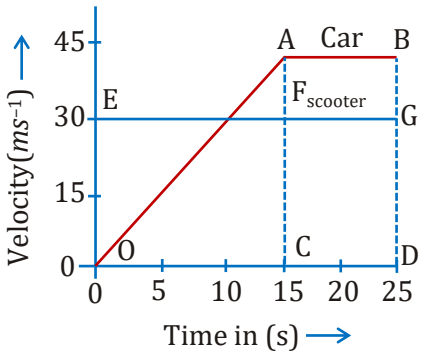
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7. During their motion, the car C_1 over takes the car C_2. But soon C_2 overtakes C_1 so determine the maximum lead that C_1 can have from C_2.

(A) 0.5m (B) 9.5m (C) 81m (D) 90m

PKN343

8. The velocity - time graphs of a car and a scooter are shown in the figure. (i) The difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively.



(A) 112.5 m and 15 s
(B) 337.5 m and 25 s
(C) 225.5 m and 10 s
(D) 112.5 m and 22.5 s

PKN344

For More Join: @IITJEE_Advanced

9. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 $km/hour$ and 72 $km/hour$, respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8 km/hr. Speed (in ms^{-1}) of this person as observed from train B will be close to : (Take the distance between the tracks as negligible)

(A) 30.5 ms^{-1} (B) 29.5 ms^{-1} (C) 31.5 ms^{-1} (D) 28.5 ms^{-1}

PKN345

10. A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2. The car will catch up with the bus after a time of :-

(A) $\sqrt{110}\,s$ (B) 15 s (C) $10\sqrt{2}\,s$ (D) $\sqrt{120}\,s$

PKN346

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	(A-P,Q,R) (B-S,T) (C-P) (D-Q)	C	D	B	A	D	B	C

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RACE
Regular Analysis Through Continuous Exercise

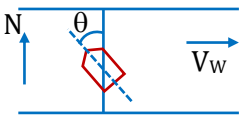
Kinematics : RACE-05

1. At a given instant, an observer stationary on the ground sees a package falling with speed v_1 at certain angle with the vertical. A pilot flying at a constant horizontal velocity relative to the ground sees the package falling vertically with a speed v_2 at the same instant. What is the speed of the pilot relative to the ground?

(A) $v_1 + v_2$ (B) $v_1 - v_2$ (C) $\sqrt{v_1^2 - v_2^2}$ (D) $\sqrt{v_1^2 + v_2^2}$

PKN347

2. A boat is to go straight across a stream that is flowing at 5.0 km/h to east. The boatman knows that his speed with respect to water is 10 km/h. The angle at which he must head the boat, and his speed relative to the shore will be :-



(A) 10 km/h at 0° (B) $5\sqrt{3}$ km/h at 60°
(C) $10\sqrt{3}$ km/h at 30° (D) $5\sqrt{3}$ km/h at 30°

PKN348

3. Two boats were going down stream with different velocities. When one overtook the other a plastic ball floating in water was seen in line with two boats. Sometime later both boats turned back simultaneously and went at the same speeds as before (relative to the water) towards the spot where the ball had been dropped. Which boat will reach the ball first?

(A) The boat which has greater velocity (relative to water).
(B) The boat which has lesser velocity (relative to water).
(C) Both will reach the ball simultaneously.
(D) Cannot be decided unless we know the actual values of the velocities and the time after which they turned around.

PKN349

4. The stream of a river is flowing with a speed of 2 km/h. A swimmer can swim at a speed of 4 km/h. What should be the direction of the swimmer with respect to the flow of the river to cross the river straight?

(A) 60° (B) 150° (C) 90° (D) 120°

PKN350

Paragraph for Question No. 5 and 6

Ram and Shyam are walking on two perpendicular tracks with speed 3 ms^{-1} and 4 ms^{-1} respectively. At a certain moment (say $t = 0$ sec) Ram and Shyam are at 20 m and 40 m away from the intersection of tracks respectively and moving towards the intersection of the tracks.

5. During the motion the magnitude of velocity of Ram with respect to Shyam, is -

(A) 1 ms^{-1} (B) 4 ms^{-1} (C) 5 ms^{-1} (D) 7 ms^{-1}

PKN351

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6. Shortest distance between them subsequently, is -

(A) 18 m (B) 15 m (C) 25 m (D) 8 m

PKN352

7. Ship A is sailing towards north-east with velocity $\vec{v} = 30\hat{\imath} + 50\hat{\jmath}\ km/hr$ where $\hat{\imath}$ points east and $\hat{\jmath}$, north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in :

(A) 4.2 *hrs*. (B) 2.2 *hrs*. (C) 3.2 *hrs*. (D) 2.6 *hrs*.

PKN353

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ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	C	D	C	D	C	D	D

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